

Macroeconomic Dynamics and Foreign Investment Flows: An Integrated Analysis and Their Impact on Nifty Returns

Dissertation

MASTERS OF ARTS IN ECONOMICS

By

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Abstract

The research analyzes the effectiveness of key macroeconomic variables on nifty returns in India. The objective of the study is to test for both short-run and long-run relationships between nifty returns and major macro variables like forex reserve, FII net flows, CPI monthly, index of industrial production, crude-oil price, gold price 10gm and rate of interest. Monthly data over the study period is considered to observe dynamic behavior of the above factors. The paper employs the Autoregressive Distributive Lag model to identify both long-run and short-run relationships while the bound testing approach is used to check for the presence of cointegration among variables. The length of optimal lag is selected using Akaike Information Criterion whereas the error correction test is used to estimate the adjustment speed towards long-run equilibrium. The findings suggest that macro-economic dynamics play a major role in determining stock market returns in India and provide useful insights for investors, policy makers and financial analysts in understanding market behavior and macroeconomic transmission mechanism.

Keywords: Nifty Returns, Macroeconomic Variables, ARDL model, Bound test, ECM, stock market

Introduction

Stock market serves as a mirror reflecting the broader economic health of a nation. When investors, policymakers, and analysts want to examine how well a country is performing financially, they often turn to market indices as a first point of reference. This connection is even more sound in developing economies, where markets tend to be more sensitive to shifting macroeconomic conditions. In countries like India, a change in inflation, currency reserves, or industrial output can spread through investor sentiment almost immediately, reshaping flow of capital and investment strategies in such a way that developed markets might absorb fast.

Despite this consequence, the Indian stock market presents a compelling case for study. The Nifty 50 index, which tracks companies across diverse sectors, has become a reliable indicator for understanding both behaviour of the market and the overall direction of the Indian economy. Though it has many pros, the question of which macroeconomic forces affect Nifty's performance and how much, remains an area important for exploring with econometric precision.

Several variables are there to shape stock market outcomes in India. Foreign exchange reserves shows the country's capacity to manage external shocks. Foreign Institutional Investor flows reflect how global capital influences India's investment climate. Inflation, as measured by the Consumer Price Index, affects purchasing power and corporate profitability. Industrial production data captures the real economy's momentum. Meanwhile, crude oil prices carry particular weight for an import oriented nation like India, while gold prices and interest rates influence both savings behaviour and the cost of borrowing.

To check on how these variables interact with Nifty returns, this paper draws on the Autoregressive Distributed Lag framework, a methodology well suited to situations where

variables exhibit mixed orders of integration, i.e. where some shows stationarity at $I(0)$ and others require 1st differencing $I(1)$ to achieve stationarity. The ARDL approach is further supported by bound testing for co-integration and an error correction mechanism that traces how the system returns to equilibrium after a disturbance. The theoretical lens through which these relationships are interpreted is the Mundell-Fleming model, which extends the classic IS LM framework to account for international capital flows, exchange rate dynamics, and cross border price effects all of which are highly relevant in today's interconnected financial environment.

The central motive of this research is therefore to empirically assess the way foreign exchange reserves, FII net flows, CPI, the index of industrial production, crude oil prices, gold prices, and interest rates collectively influence Nifty returns. The paper is structured to build this argument progressively, moving through a review of existing literature, a detailed explanation of data and methodology, a presentation and discussion of empirical findings, and finally, a set of conclusions alongside actionable policy recommendations.

Research Question:

The gradual integration of the Indian economy with international financial markets along with rising influence of macroeconomic indicators has brought many issues forefront that pertain to the drivers behind stock market performance. Keeping the above mentioned scenario in view, this research paper aims to explore the central research query related to the impact of macro-economic variables and capital flows on Nifty returns in Indian scenario. More precisely, this paper will seek to find out if macroeconomic indicators like inflation, foreign exchange reserves, foreign institutional investment, crude oil prices, gold prices, industrial production, and interest rates have any influence on Nifty returns in the short term as well as the long term. It aims at investigating the widespread relation between macro-economic indicators and fluctuations in stock market using econometric techniques.

Literature Review:

In the Indian context, several studies find that inflation, exchange rate and interest rate related variables frequently explain variation in Sensex/Nifty returns, but results are mixed across time periods, variable choices and methods (cointegration/VECM, VAR, Granger causality, regression, ARCH-family models). Recent Indian work (Amit Headu, JIER 2024) also stresses the role of international indices (Dow Jones) and exchange-rate movements alongside domestic macro variables. A recurring finding in the India literature is that foreign institutional inflows (and, to a lesser extent, FDI) matter for liquidity, volatility and short to medium term price movements. Work on cross sectional return patterns (value vs growth, size effects) suggests that apparent excess returns can reflect compensation for specific risks (leverage, earnings uncertainty, dividend cuts) rather than pure mispricing. Chen(1998) and related studies show value premiums vary across countries and are linked to firm distress and market growth regimes- implying that macro/market structure differences matter for return patterns. This complements macro-level studies by linking firm-level risk structure to aggregate return formation.

The connectivity between macro-economic variables and performance of stock market has been examined in both developed and emerging economies. Researchers argue that stock market indices reflect the overall economic health of a country and are sensitive to changes in the key macro-economic indicators like inflation, rate of interest, exchange rates, foreign investment flows and commodity prices.

One of the pioneering studies in this area was conducted by Chen, Roll and Ross, 1986, who investigated the indulgence of macroeconomic forces on market returns in the United States. Using an arbitrage pricing framework, it is observed that variables such as industrial output, inflation and the growth spread between long and short term government bonds significantly describes stock market returns. Their findings provided strong empirical evidence that macroeconomic variables play an important role in determining equity prices.

Extending this line of research to Asian markets, Mukherjee and Naka, 1995, examined the dynamic interlinkage between macroeconomic variables and price of stocks in the Japanese stock market using a vector error correction mechanism. Their study showed that money supply and foreign exchange reserves possess a long run equilibrium relationship with price of stocks, whereas exchange rates did not show a significant long-term association. This study highlighted the importance of monetary variables in influencing stock market movements in emerging economies.

Focusing on the Indian context, Pethe and Karnik 2000 analyzed the interaction between Indian stock market indices and macroeconomic variables during the post liberalization period. Their results indicated that the Indian stock market exhibited weak-form efficiency, implying that stock prices were influenced by macroeconomic changes but did not fully incorporate all available information instantaneously. The study emphasized that macroeconomic variables and stock prices in India are interconnected, though the strength of the connectivity varies across time.

Further evidence from developed markets was provided by Ratanapakorn and Sharma, 2007, who examined both short run and long run relationships between macroeconomic variables and the U.S. stock market index, S&P 500. Their findings suggested that there exist a negative connection between long term interest rates and stock prices, while other macroeconomic variables such as supply of money and industrial output showed a positive association with market performance. This study reinforced the idea that macroeconomic fundamentals significantly influence equity markets.

More recent studies have shifted focus toward external and global macroeconomic factors, particularly for emerging economies like India that are highly dependent on foreign capital flows, crude oil imports, and exchange rate stability. Variables like foreign institutional investment, crude oil prices, gold prices, balance of payments, and foreign exchange reserves have gained prominence in explaining stock market behaviour. However, empirical findings remain mixed, with some studies reporting significant relationships while others finding limited or weak associations.

Overall, the existing literature suggests that although macroeconomic variables influence stock market performance, the magnitude and direction of their impact differ across countries, time periods, and selected variables. Moreover, the Indian stock market has often been found to exhibit weak-form efficiency, indicating that not all macroeconomic information is fully

reflected in stock prices. This inconsistency in empirical evidence highlights the need for further research focusing on recent periods and incorporating relevant external macroeconomic factors to better understand their impact on the NSE Nifty 50.

The linkage among macroeconomic variables and performance of stock market has been a central theme in study of economics in finance, particularly in rising markets like India where economic fundamentals and market dynamics evolve rapidly. Numerous empirical studies have examined how macroeconomic indicators influence stock market movements, market efficiency, and investor behaviour.

Recent evidence by Naik, 2024 highlights that the Indian equity market is significantly driven by macroeconomic indicators such as Foreign Portfolio Investment, Index of Industrial Production, Consumer Price Index, and exchange rates. The study emphasizes that fluctuations in these variables shape investor sentiment and market outcomes. However, it also points out that the connectivity between GDP and performance of stock market remains insufficiently explored, leaving scope for deeper investigation.

Similarly, Hedau 2024 examined the effect of global economic factors on the Indian stock market and found that movements in the Dow Jones Industrial Average and exchange rate fluctuations exert a considerable influence on the NIFTY 50 index. The study underscores the importance of international linkages but notes that political stability and bilateral relations are often overlooked, despite their potential impact on market volatility and investor confidence.

Focusing on stock price volatility, Ranjith Kumar and Banu, 2023 analysed the impact of key macroeconomic variables on the Indian stock market using advanced statistical techniques. Their findings confirmed that macroeconomic variables significantly push stock market dynamics. However, the study acknowledged limitations such as the exclusion of interest rate volatility and political factors, which could further explain variations in stock prices.

An extensive assessment of market efficiency was carried out by Siddhpuria and Manani 2023, who investigated the influence of multiple macroeconomic variables—including CPI, FII flows, repo rate, CRR, SLR, exchange rate, and prime lending rate—on major Indian indices such as Nifty, Bank Nifty, and Sensex. Employing time series techniques, cointegration analysis, and Vector Error Correction Models, the study found significant long run and short run relationships between macroeconomic variables and stock market indices, reinforcing the notion that Indian markets are influenced by economic fundamentals.

Long-term empirical evidence is provided by Marvadi, 2022, who evaluated the impact of macroeconomic indicators on Indian stock market performance during the post-liberalization period from 1993–94 to 2018–19. The study shows conclusion that that gross domestic product and foreign exchange reserves significantly explained variations in the Sensex, while variables such as foreign exchange reserves, crude oil prices, and currency depreciation were key drivers of NSE Nifty fluctuations. Nevertheless, the study highlighted the limited inclusion of monetary variables such as broad money supply.

Earlier research by Das and Megaravalli, 2017 explored the connection between macroeconomic variables and Indian stock market indices using quarterly data and econometric tools such as Johansen cointegration and granger causality tests. Their findings detects a positive correlation between macroeconomic indicators and stock market indices, along with proof of causality running from the nifty 50 index to exchange rates and call

money rates. However, the study was constrained by a limited time frame and a narrower selection of macroeconomic variables.

Overall, the reviewed literature establishes that macroeconomic variables such as gross domestic product, foreign exchange reserves, inflation, exchange rates, and global economic indicators significantly influence Indian stock market behaviour. However, the direction and magnitude of these relationships vary across studies, time periods, and methodologies. Moreover, several studies report moderate or weak-form market efficiency, suggesting that macroeconomic information is not always fully or immediately reflected in stock prices. These inconsistencies underline the need for comprehensive studies using extended time horizons, multiple econometric techniques, and a broader set of macroeconomic variables—particularly GDP and foreign exchange reserves—to better understand the efficiency and dynamics of the Indian stock market.

Data & Methodology

Methodology:

This working uses a quantitative research design using secondary monthly time series data spanning the last 26 years. Nifty returns are taken as the dependent variable, while key macroeconomic variables such as exchange rate, inflation, rate of interest and foreign investment indicators serve as independent variables. All data are collected from reliable sources such as NSE, RBI, MOSPI, and international financial databases. Given results of cointegration, the study applies Autoregressive Distributive Lag to capture adjustments between Nifty returns and macroeconomic variables, and foreign investment inflows. The ARDL framework also helps to identify the adjustment speed towards equilibrium.

Data:

We conduct the econometric analysis of the study on the basis of a monthly time series with an observation period ranging from February 1999 to December 2025, amounting to a total of 323 observations. The purpose of preparing the dataset is to investigate the series of macroeconomic variables that affect stock market performance. The dataset includes several important macro financial and stock market performance indicators for the Indian economy.

It is a time series data where each observation represents the last trading day of each month. The variables have been taken on a monthly basis to test for short-run fluctuations and the dynamic interactions between financial markets and fundamentals.

The dataset has the following variables:

Date – Monthly time index ranging from 1999 to 2025.

NIFTY_PRICE - The monthly closing value of nifty index.

NIFTY_RETURNS- Monthly return of percentage calculated from Nifty data.

CPI monthly – Consumer Price Index, representing inflation dynamics.

Crude Oil Price – International crude oil prices that include global commodity price shocks.

FOREX_RESERVE – India's foreign exchange reserves, which reflect the external sector position.

FII_NET_FLOWS – This variable represents the net flow of foreign institutional investment based on global investors' sentiments.

IIP (cent). – Index of Industrial Production Growth Rate Percent, a proxy for real economic activity.

Gold Price (10gm) – Domestic gold price per 10 grams, an alternative investment option and hedge against inflation.

Interest Rate – This variable represents the percentage change in interest rate over the time period.

The dataset has 323 observations for the base stock market-related variables (NIFTY price and returns). The stock exchange index of NIFTY has shown substantial returns during the selected period of time, with the minimum trade value at about 914 and the maximum at about 26,203, indicating the Indian stock market growth in the long term. Additionally, the average monthly returns are 1.02%, with considerable volatility in the market in boom and subsequent crises, indicated by the standard deviation of 6.31%.

Inflation, calculated by using the CPI, indicates a mean of 212.61 with a significant level of variability, reflecting the change in the price structure over time. Variability in crude oil prices, on the other hand, shows significant variability in the mean price of crude oil, i.e., around 62.38 USD, with a range of prices from 12.31 to 139.96.

Foreign exchange reserves have shown a strong upward trend, with a rise from approximately 118,062 million USD to over 5.17 million USD, showing stronger resilience in the external sector. FII net flows have also shown strong volatility, with its value fluctuating from high negative numbers to positive numbers.

Here, the average growth of industrial production IIP is 5.34%, but it undergoes sudden contractions during periods of economic recession. The minimum growth is reported to be -16.6% . In the case of gold prices, there is a high growth trend. The mean value is pegged at 24,947 rupees per 10 grams of gold.

Model

The theoretical framework of the paper is based on the Mundell Fleming model of the open economy. The Mundell Fleming model is extensive form of the IS LM framework to include the results of the external sector through the balance of payments. The Mundell Fleming model is an open-economy model of monetary policy, which incorporates the effects of capital mobility and exchange rate flexibility. In India, especially over the period of post-liberalization (1999 to 2025), foreign capital inflow, stability of the macro-economy, and the monetary sector have been very important variables to affect the financial sector of the economy. This paper is an expansion of the traditional Mundell-Fleming model to include the financial sector's performance, represented by Nifty returns.

Under the Mundell Fleming model, the equilibrium of the goods market is expressed by the IS curve and will be influenced by income or output and interest rates and net exports. The equilibrium of the money market is represented by the LM curve and will be influenced by income and rate of interest. The balance of payments is expressed by the balance of payment curve and will be influenced by interest rate differentials and world conditions. Foreign investment will shift the curve outward if raised, resulting in surplus in the capital account and appreciation of the currency. The central bank will attempt to stabilize the value of the currency from the effect of the increase in the forex reserves. This increase in forex reserves will, in turn, increase the level of liquidity in the economy, reducing interest rates and hence investing, improving corporate earnings, and increasing the overall returns from the stock market. Hence, the foreign investment and the forex reserves are expected to be positive factors for the returns on the Nifty index.

The standard Mundell–Fleming model consists of three equations:

IS curve, goods market equilibrium

$$Y = C(Y-T) + I(r) + G + NX(e, Y, Y_1)$$

Where:

- Y = output
- r = rate of interest
- e = rate of exchange
- NX = net exports

LM curve, money market equilibrium

$$M/P = L(Y, r)$$

Balance of Payment curve

$$BP = CA(Y, e) + KA(r - r_1)$$

Where:

- CA = current account
- KA = capital account
- r_1 = world interest rate

We extend this framework by incorporating financial markets explicitly. In modern open economies, stock markets react immediately to macroeconomic shocks and capital flows.

We define:

$$NR_t = f(FII_t, FR_t, OIL_t, GOLD_t, IIP_t, INF_t)$$

Where:

- NR_t = Nifty returns
- FII_t = Foreign investment inflows
- FR_t = Forex reserves
- OIL_t = Crude oil prices
- $GOLD_t$ = Gold prices
- IIP_t = Industrial output (proxy for output Y)
- INF_t = Inflation

Industrial output, represented by the index of industrial production, acts as a proxy for real economy within the context of the IS model. An increase in industrial output implies improved productivity, higher corporate profits, and stronger demand, which raises stock market performance. On the other hand, stock returns fall due to various factors related to inflation. An increase in inflation raises uncertainty, reduces real purchasing power, and leads to upward pressure on nominal interest rates due to a higher discount factor for stocks of future earnings. This leads to a decrease in Nifty returns.

The crude oil price enters the model in terms of external sector. Since India is a net crude oil importer, higher crude oil prices will be responsible for current account deficit to spread, thereby affecting production costs and driving inflation, factors that negatively impact stability and corporate profits, leading to decelerated stock exchange returns. Similarly, the price of gold enters as a substitute asset variable; the price of gold has often been considered a refuge asset during times of uncertain shocks, such that higher or rising asset prices signal risk aversion, resulting in negative NIFTY returns.

In accordance with the extended Mundell-Fleming model, the reduced form theoretical model can be written as follows: returns on the Nifty as a function of foreign investment inflows, foreign exchange reserves, industrial output, inflation, crude oil prices, and gold prices. According to the model, in the Indian context, Nifty returns would exhibit a positive association with foreign investment inflows, foreign exchange reserves, and industrial output. Similarly, it would show a negative association in relation to inflation, crude oil prices, and gold prices. Empirically, in furtherance of the above theoretical constructs, the model will be tested by incorporating monthly data from the year 1999 to 2025 through various methodologies of time-series econometric analysis. Accordingly, it can be noted that the overall model bridges macroeconomic theory in an open economy and financial markets. The study will establish a single framework to analyze macroeconomic and foreign capital contributions to financial markets.

We can express the extended reduced-form model as:

$$NR_t = \alpha + \beta_1 FII_t + \beta_2 FR_t + \beta_3 IIP_t + \beta_4 INF_t + \beta_5 OIL_t + \beta_6 GOLD_t + \epsilon_t$$

Where:

- $\beta_1 > 0$
- $\beta_2 > 0$
- $\beta_3 > 0$
- $\beta_4 < 0$
- $\beta_5 < 0$
- $\beta_6 < 0$

The equation represents the financial-market-augmented Mundell–Fleming model.

Empirical Approach

The empirical investigation in this study is performed using time series data on the returns of the NIFTY index and chosen macroeconomic variables, which include foreign exchange reserves, net FII flows, consumer price index, index of industrial production, price of crude oil, gold price 10gm, and rate of interest. The sample size is adequately large to analyze the inter relationship between stock market returns and macroeconomic factors in India. Nifty returns and macroeconomic factors are considered as time series data, and their behavior in terms of statistics is quite complex in the sense that they might be exhibiting trends, volatility, or even non stationarity. Hence, it is imperative that an appropriate econometric model be used for this purpose.

In order to accomplish the purpose of investigating the impact of macroeconomic variables on nifty returns, the Autoregressive Distributed Lag model is adopted in the present research study. The ARDL method has proven its efficiency when applied to time series data, especially when the variables under consideration consist different order integration, which means a mixture of I (0) and I (1) variables. Contrary to other cointegration methods such as Johansen or Engle Granger approaches, the ARDL model is capable of estimating both short run and long run relationships simultaneously. Thus, the adoption of the ARDL model becomes logical for testing the impact of macroeconomic factors on the stock return in the Indian stock market.

Prior to conducting the regression model for ARDL, many tests have been performed in order to validate the model. These include descriptive analysis to understand nature of data, correlation result to examine the initial relationship between variables, unit root test to check the stationarity of the series, selection of the optimum lag length in order to identify the suitable lags, and bounds testing for ARDL model to identify long-run cointegration.

✓ **Descriptive Statistics:**

For this analysis, the data set utilized is based on 323 months of observation period and 11 variables. This indicates that a large enough time series data set has been chosen for conducting an econometric analysis of the data. All macro and financial variables are expressed in double precision format. This is done in order to ensure more accurate estimates and avoid any possibility of rounding errors. The combination of a string date variable and the numeric time variable in the data set indicates that the data is properly organized for time series analysis and can be set as a time variable for the purpose of analysis using models such as ARDL and others. The average monthly return on NIFTY is 0.0102 (or 1.02%), while its standard deviation is 0.0631. This shows that there was moderate fluctuation in stock market returns. The range between the minimum and maximum values, namely -0.3066 to 0.2473 , implies that there were drastic falls as well as gains in the stock market. Consumer Price Index is characterized by mean 212.61 and standard deviation 148.81, which demonstrates huge fluctuations in the economy due to rising levels of inflation during the period under consideration. The CPI varies between 42.97 and 532.59. The crude oil price is 62.38 with a standard deviation 26.19 and variations between 12.31 and 139.96, which is related to the fluctuations in global crude oil price and may affect the cost of production for the Indian economy. Foreign exchange reserves exhibit very high levels of mean (1,800,629) with a large deviation of 1,407,399, thereby demonstrating a high level of reserves accumulated in the foreign reserve account of India. The net flow of foreign institutional investors stands at 5513.22 with a large standard deviation 9024.52 and a minimum of -10186.65 . The mean index of industrial production is 5.34 with a standard deviation 9.09; therefore, there is considerable growth in industrial production but with some instances of extreme deviations in terms of maximum and minimum values, i.e., 134.4 and -16.6, respectively. The average interest rates are 6.61 with a relatively small standard deviation of 1.33, signifying consistent monetary policy changes over the sample period, from minimum to maximum interest rates, i.e., 4.25 to 10.25. The gold prices' mean value is 24,946.98, while the standard deviation is 19,521.93, indicating considerable upward trend and volatility in gold prices, supporting the hypothesis that gold is a safe investment. Overall, the findings reveal that there is enough variance and volatility within the selected data set, making the use of time series modeling through the ARDL methodology feasible.

✓ **Regression Results:**

The multiple regression models analyzes the influence of foreign exchange reserves, FII net inflows, CPI, IIP, crude oil price, gold price, and interest rate on NIFTY returns. According to the results of the regression, the entire model has low statistical significance based on the F-statistic, which equals 0.68 and a probability value of 0.6925, which is considerably greater than the 5% significance level. In other words, the set of macroeconomic factors selected as independent variables has no significant

effect on nifty returns in the short run. In addition, the R squared value equal to 0.0184 implies that the macroeconomic factors considered explain only 1.84% of the variance of NIFTY returns, whereas the rest of the variation depends on numerous other factors including investors' sentiments and expectations, global financial conditions, and government policies. The adjusted R squared is negative (−0.0088), meaning that including many independent variables into the model does not add value to the explanation of NIFTY returns. Finally, the root mean squared error equals 0.06852.

Table 2: Regression

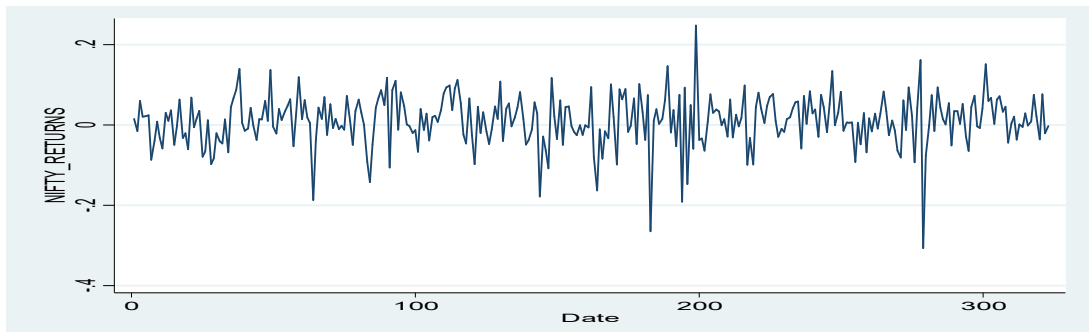
OLS Regression Results Dependent Variable: NIFTY_RETURNS							
Model Fit Statistics							
Number of Obs	F(7, 253)	Prob > F	R-squared	Adj R-squared	Root MSE		
261	0.68	0.6925	0.0184	-0.0088	0.06852		
Variable	Coefficient	Std. Error	t-statistic	P> t	95% CI Lower	95% CI Upper	
FOREX_RESERVE	-2.72E-09	2.03E-08	-0.13	0.894	-4.27E-08	3.73E-08	
FII_NET_FLOWS	0.00000114	0.000000654	1.75	0.082	-0.000000146	0.00000243	
CPImonthly	0.0000296	0.0000524	0.57	0.572	-0.0000736	0.0001329	
IIPcent	0.0004538	0.0004699	0.97	0.335	-0.0004716	0.0013793	
CrudeOilPrice	-0.0000857	0.0001908	-0.45	0.654	-0.0004615	0.0002901	
GoldPrice10gm	-2.32E-08	0.00000161	-0.01	0.989	-0.00000319	0.00000314	
Interestrates	-0.0017578	0.0043151	-0.41	0.684	-0.0102559	0.0067402	
_cons	0.0177748	0.0341309	0.52	0.603	-0.049442	0.0849917	
Note: Green shading = statistically significant at 5% level. Coefficients shown in scientific notation where applicable.							

The regression output shows that the individual coefficients for the variables include positive FII net flows with a value 1.14e-06 with a p-value 0.082. The positive coefficient implies that a positive correlation exists between the variable and nifty returns, although the significance level (p-value) is low (10%), showing a significant effect. It also shows that an increase in foreign capital investment will increase the returns. The negative coefficient for forex reserves shows that the variable has little effect on stock returns, but the significance level (p-value) is extremely high (0.894). It suggests that the forex reserve is used to promote long-term economic development rather than influencing the stock returns. The coefficient for the CPI index is positive and insignificant (p-value 0.572), indicating that the stock returns are not affected by inflation. The coefficient for industrial production index is also positive (0.0004538) but insignificant (p-value 0.335). It shows that the growth of production does not immediately generate stock returns. The coefficient for crude oil price is negative (−0.0000857).

✓ Trend Graph:

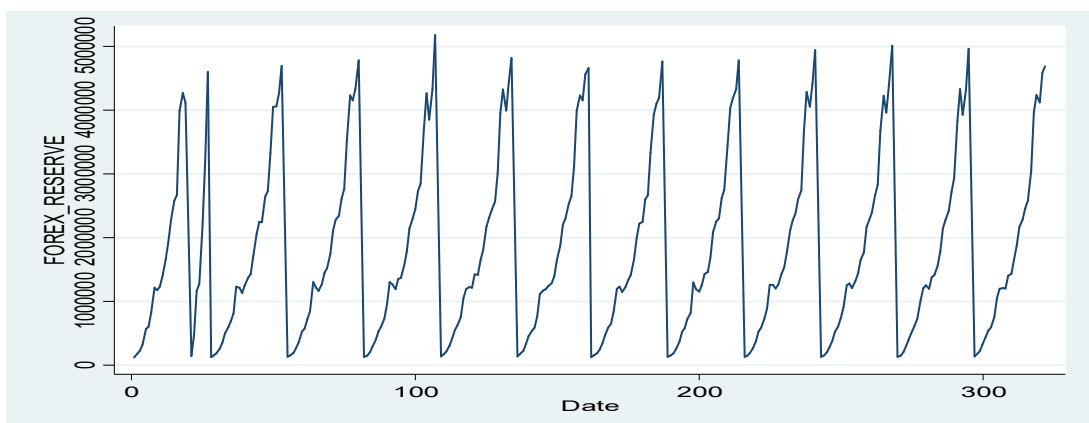
The trend graph is plotted after regression to visually examine time wise movement and pattern of NIFTY returns and macroeconomic variables, which helps in understanding whether the variables exhibit trends, volatility or structural change over the study period. This visual inspection supports the regression analysis by providing a clearer picture of data behavior and helps in identifying non stationarity, fluctuations and dynamic relationships.

Nifty Returns



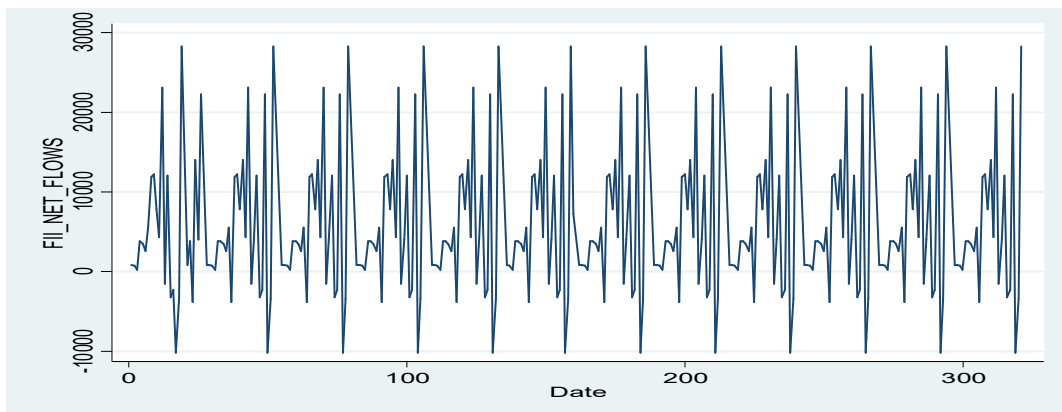
From the given graph, there are fluctuations in the return on investment on NIFTY, which clearly indicates the volatile nature of the Indian stock market. There are times when there is a very steep increase as well as a rapid fall, signifying the volatility of nifty returns, especially in terms of its reactions to various economic factors and changes taking place around the world.

Foreign Exchange Reserve



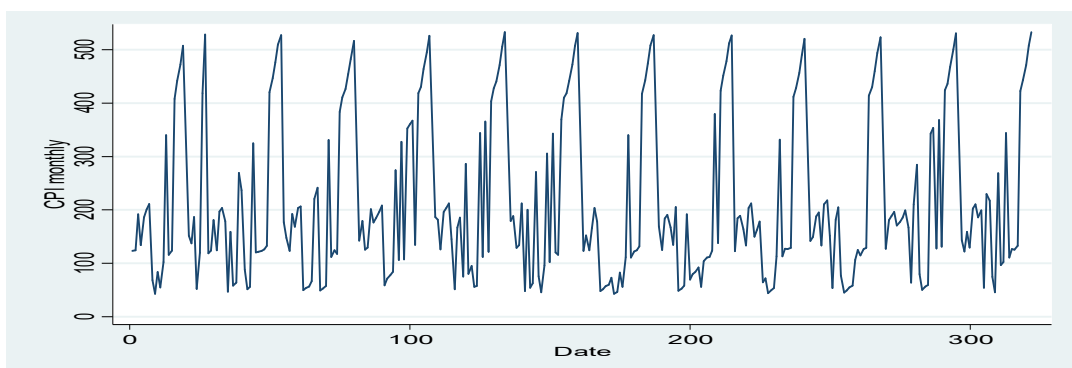
The pattern chart on foreign exchange reserves reveals an upward trend throughout the analysis period with certain fluctuations, reflecting the increasing amount of foreign exchange reserves in India's economy. This signifies an improvement in external stability and better management of foreign exchange reserves by the central bank, but this is affected by short-term movements in the international environment.

FII Net Flows



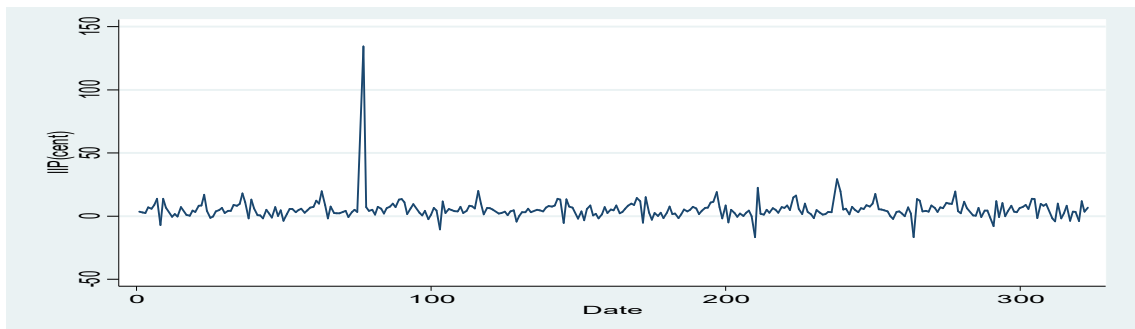
The chart of trends in net FII inflows is characterized by considerable volatility with many sudden upward and downward spikes during the study period. These points to volatility of flows of foreign institutional investments on nifty returns. The recurring pattern of both inflows and outflows reveals that foreign investors react dynamically to changing economic and political circumstances in the global economy. On balance, the chart clearly demonstrates that FII flows are volatile and may affect the performance of nifty return in the short run.

Consumer Price Index



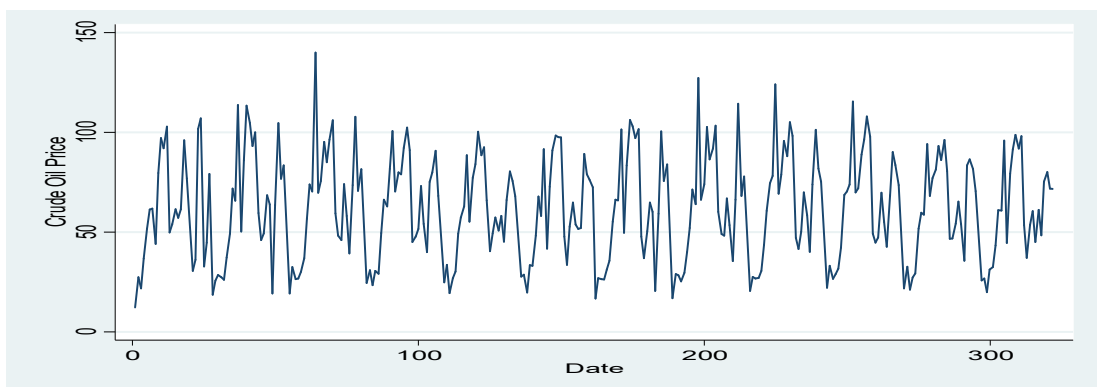
The graph depicting the trend in the CPI depicts a consistent trend with fluctuations. This implies that the level of inflation is on the increase throughout the period under study. This means there is an increasing rate of prices over time due to changing demand supply trends and monetary policies. The entire trend depicted is therefore inflationary.

Index of Industrial Production



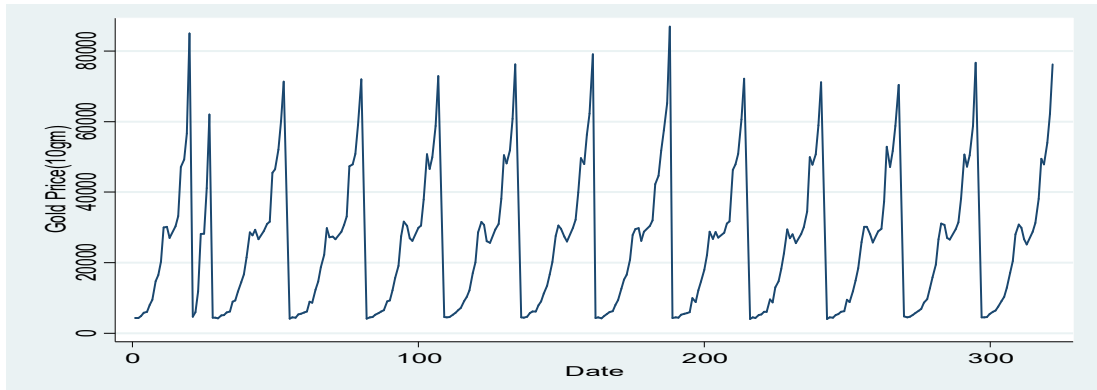
The chart on IIP is evident to have some kind of fluctuations, where there are instances of rapid increase and decrease. Thus, it implies that there have been many ups and downs within the period under discussion regarding industrial production growth. Hence, it can be concluded that the growth in industrial production has been unstable, depending upon economic cycles, economic conditions, and policies.

Crude Oil Price



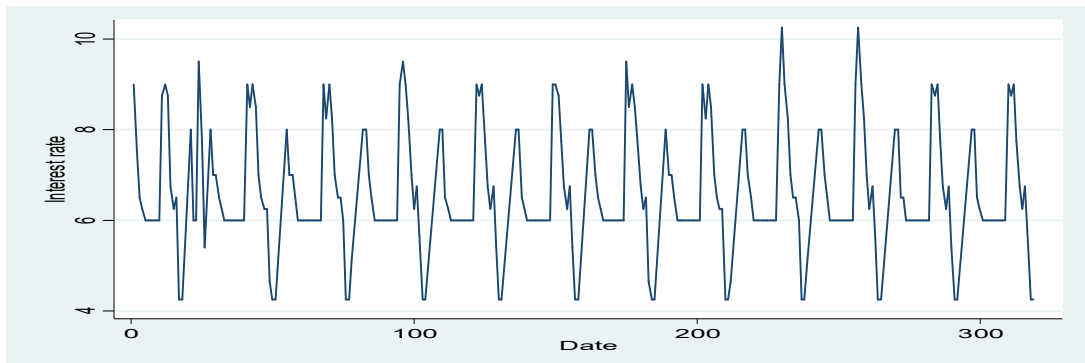
The graph displays that oil price is highly volatile over the period of study with significant and rapid fluctuations. There is no discernible long term trend; instead the data points show frequent and sharp short term variations. The pattern suggests that the variable is subject to numerous short term influences rather than a single, dominant long term driver.

Gold Price



The graph on gold price trend exhibits a highly cyclical pattern where the price constantly experiences sharp increases followed by a sudden drops at regular intervals. This suggests potential regular market interventions or highly predictable short term volatility rather than a smooth long term trend.

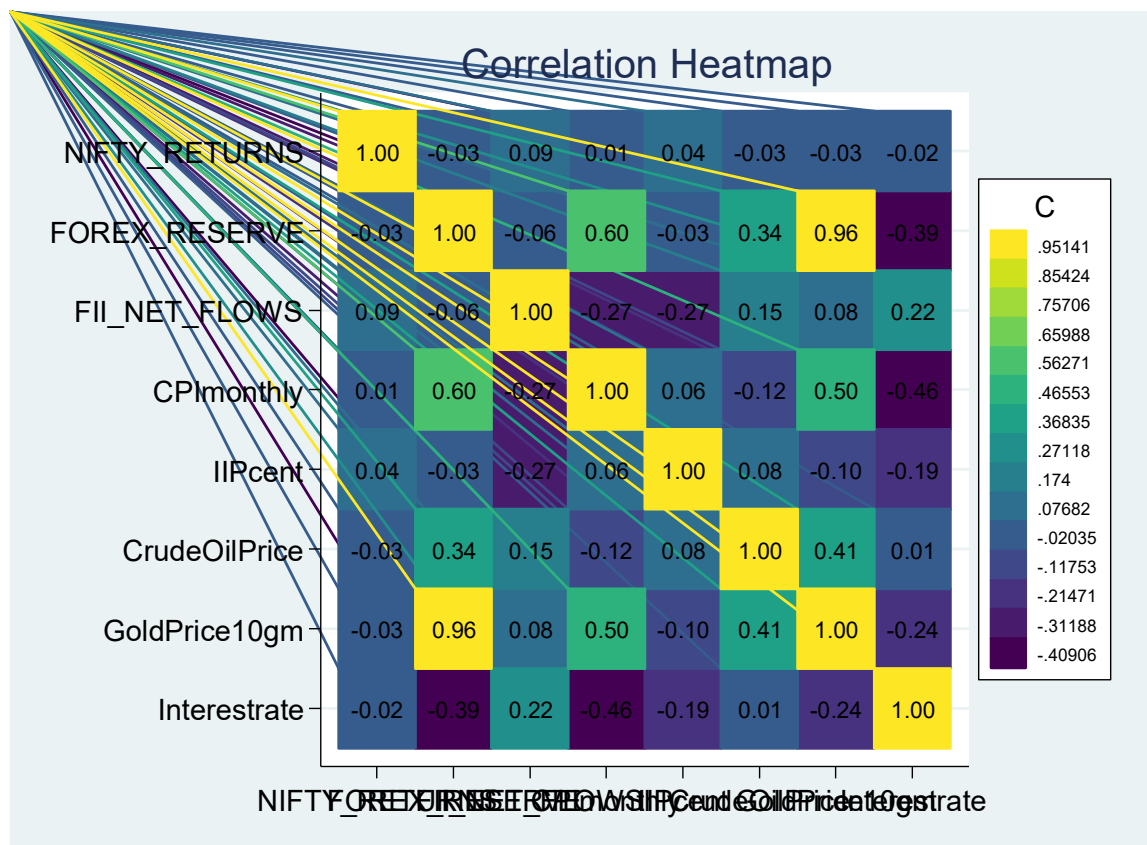
Interest Rate



The trend graph of interest rate illustrates that interest rate fluctuates significantly over the time period, with rates generally staying within the range of 4% to 10% without a clear long term upward or downward trend.

✓ Correlation Heat map

To analyze the relation between the variables under study is an important test to fulfill our objective of study. It indicates that nifty returns have very low correlation with macroeconomic variables and signifies very little linear relationship between nifty returns and the economic variables under consideration. We can see that nifty returns are slightly positively related with FII net flows (0.0910) and index of industrial production (0.0382). Though very low negative correlation has been observed between nifty returns and the variables forex reserve (-0.0291), crude oil price (-0.0331), gold price (-0.0257). This implies a very little direct influence of macro variables on stock market.



Again if we check correlation among the independent variables, it is observed that a very high correlation exist between the variables forex reserve and gold price (0.9635) and moderately high correlation between forex reserve and consumer price index CPI (0.5984). These results show evidence of existence of multicollinearity between the variables. Economically as forex reserve increase, gold price tend to rise automatically because gold is considered to be an important component on country's foreign reserve and central bank uses gold holdings as part of risk management strategies.

✓ Augmented Dickey Fuller Test

Checking for stationarity is an important step for analyzing the nature of the variables. The best way is to use Dickey Fuller Test. As the data set comprises time series variables, it may be possible that the variables could have trends and unit roots, which will result in spurious regression. It is necessary to conduct the Dickey Fuller test to find out whether the variables are stationary at level or should be differenced. This will ensure the proper conduct of the econometric analysis and check whether the variables are appropriate to apply the ARDL method, where I (0) and I (1) variables are used but not I (2) variables.

Table 4: ADF test

Variable	Lag	T Statistic	P Value	1% critical value	5% critical value	10% critical value
Nifty Returns	0	-11.910	0.0000	-3.454	-2.877	-2.570
Forex Reserve	0	0.325	0.9785	-3.457	-2.878	-2.570
	1	-12.806	0.0000	-3.458	-2.879	-2.570
FII Net Flow	0	-16.120	0.0000	-3.458	-2.879	-2.570
CPI monthly	0	-3.226	0.0185	-3.457	-2.878	-2.570
IIP cent	0	-14.381	0.0000	-3.457	-2.878	-2.570
Crude Oil Price	0	-8.140	0.0000	-3.457	-2.878	-2.570
Gold Price 10 gram	0	-1.333	0.6141	-3.459	-2.880	-2.570
	1	-12.027	0.0000	-3.464	-2.881	-2.571
Interest Rate	0	-12.806	0.0000	-3.458	-2.879	-2.570

The table shows that nifty returns, FII net flows, CPI monthly, IIP, Crude Oil price and Interest rate all have p value less than 0.05 which implies they don't have unit root and all are stationary at I (0). So, there's no need of differencing. Only Forex Reserve and Gold Price are showing p-value greater than 0.05 implying they possess unit root. So Dickey Fuller test at first difference shows that these variables are stationary at me (1).

✓ Auto Regressive Distributive Lag (ARDL)

The dataset shows mixed integration I (0) and I (1), the best model is Auto Regressive Distributive Lag because this model allows I (0), I (1), mixed variables, short run and long run analysis and avoid spurious regression. To determine optimal lag length we used Vector Auto Regression lag selection criteria that best explains the data without overfitting. The target is to choose the lag where information criteria are minimum. The VARSOC test shows Akaike Information Criteria is minimum at lag 2. It provides best balance between model accuracy and complexity. For a good model fit, AIC is chosen because it works well in macroeconomic and financial time series data and captures dynamic relationships better. The result shows that Nifty Returns are influenced by past two periods of macroeconomic variables making lag 2 optimal choices to capture dynamics in short run and the model gets stabilized. To check the effect of the macroeconomic variables on Nifty in short run dynamics we run ARDL model.

Table 6: ARDL test

ARDL Results Dep. Var: NIFTY_RETURNS							
A	B	C	D	E	F	G	H
ARDL Results Dep. Var: NIFTY_RETURNS							
PART 1: ARDL(1,2,0,0,0,0,0) — Standard (without ec option)							
Model Fit							
Obs	F(10,183)	Prob > F	R-squared	Adj R-squared	Root MSE		
194	1.83	0.0574	0.0911	0.0414	0.071		
Variable	Coef.	Std. Err.	t	P> t	95% CI Lower	95% CI Upper	
L1. NIFTY_RETURNS	0.0327308	0.0738824	0.44	0.658	-0.1130401	0.1785016	
FOREX_RESERVE (--)	-7.62E-08	3.57E-08	-2.14	0.034	-0.000000147	-5.78E-09	
L1. FOREX_RESERVE	0.00000015	4.89E-08	3.07	0.002	5.35E-08	0.000000246	
L2. FOREX_RESERVE	-8.24E-08	4.36E-08	-1.89	0.061	-0.000000169	3.69E-09	
FII_NET_FLOWS	0.00000238	0.000000832	2.86	0.005	0.000000737	0.000000402	
CPImonthly	0.0000499	0.0000676	0.74	0.461	-0.0000834	0.0001832	
IIPcent	0.001348	0.0011089	1.22	0.226	-0.0008399	0.0035359	
CrudeOilPrice	-0.000189	0.0002461	-0.77	0.443	-0.0006747	0.0002966	
GoldPrice10gm	0.00000034	0.00000217	0.16	0.876	-0.00000394	0.00000462	
Interstrate	-0.0032622	0.00554	-0.59	0.557	-0.0141927	0.0076683	
_cons	0.0177464	0.0422355	0.42	0.675	-0.0655847	0.1010774	

The results of ARDL model shows R squared value is 0.091 i.e., 9.1% variation in Nifty is explained by these variables. P-value and F stat are significant at 10% level. So, the model is acceptable as it is seen that the variables have short run significant impact on Nifty returns. CPI, IIP, Interest rate, Crude oil price and gold price shows weak short run effect. Initially Forex reserve increase reduces nifty returns in short run. Higher forex reserve may signal currency intervention, macroeconomic tightening and reduced market liquidity. At lag 1 it becomes significant implying forex reserve increases nifty returns after one period. So it shows a dynamic relationship with nifty returns indicating a delayed macroeconomic influence on stock market. FII net flows shows that it increases nifty returns in short run which implies it improves liquidity, market demand, stock prices and investors' confidence. This supports capital flow theory and macro finance linkage.

✓ Bound Test & Error Correction Mechanism

In order to check whether these variables are cointegrated in long run or not, we perform ARDL bound test. Bound test requires error correction mechanism because we need to check the short run and long run impact on nifty returns along with its speed of adjustment.

For hypothesis testing we take

H_0 : no long run relationship (no cointegration)

H_1 : long run relationship exists (cointegration)

If the F stat value is greater than the upper bound then the null hypothesis is rejected and it implies cointegration exists between the variables in long term.

Table 7: Bound Test

A1	Pesaran, Shin & Smith (2001) Bounds Test for Level Relationship H0: No Level Relationship									
	A	B	C	D	E	F	G	H	I	J
1	Pesaran, Shin & Smith (2001) Bounds Test for Level Relationship H0: No Level Relationship									
2	Test Statistics									
3	F-statistic	25.784	Case	Case 3						
4	t-statistic	-13.092	k (regressors)	7						
6	Stat	10% I(0)	10% I(1)	5% I(0)	5% I(1)	1% I(0)	1% I(1)	p-val I(0)	p-val I(1)	
7	F	2.055	3.185	2.367	3.573	3.035	4.387	0	0	
8	t	-2.554	-4.242	-2.86	-4.589	-3.457	-5.244	0	0	
10	Decision at Significance Levels									
11	10%	Reject H0 (.r)	5%	Reject H0 (.r)	1%	Reject H0 (.r)				
12										
13	Kripfganz & Schneider (2020) critical values. Decision: Reject H0 at all levels → long-run cointegrating relationship exists.									
14										

Here the value of F-stat is 25.784 which is greater than the upper bound (I (1)) i.e., 4.387. So the null hypothesis is rejected at all significant level. This implies that macroeconomic variables and capital flows will move together with Nifty performance in the long run. Even when there are deviations in the short run, they will revert to equilibrium.

To confirm stability of the model in long run we perform ECM test. Economically if the macroeconomic shocks disturb nifty returns, ECM shows how fast the stock market adjusts and corrects the shocks. It measures the speed at which deviations in short run adjust back to equilibrium in long run. A negative and significant ECT coefficient confirms long run convergence. Once cointegration is established, ECM turns out to be the appropriate form of ARDL and both of these models do not go hand in hand. The relationship between variables is discovered through ARDL whereas ECM is used for interpreting the process of adjustment.

Table 7: ECM in ARDL

ARDL(1,2,0,0,0,0,0) Regression Dep. Var: D.NIFTY_RETURNS ec option								
	A	B	C	D	E	F	G	
1	ARDL(1,2,0,0,0,0,0) Regression Dep. Var: D.NIFTY_RETURNS ec option							
2	Model Fit							
3	Obs	R-squared	Adj R-squared	Log-likelihood	Root MSE			
4	194	0.5531	0.5287	243.597	0.071			
6	Term / Variable	Coef.	Std. Err.	t	P> t	95% CI Lower	95% CI Upper	
7	ADJ — NIFTY_RETURNS L	-0.9672692	0.0738824	-13.09	0	-1.11304	-0.8214984	
8	LR — FOREX_RESERVE	-0.00000009	3.28E-08	-0.27	0.784	-7.37E-08	5.57E-08	
9	LR — FII_NET_FLOWS	0.00000246	0.000000891	2.76	0.006	0.0000007	0.00000422	
10	LR — CPImonthly	0.0000516	0.00007	0.74	0.462	-0.0000865	0.0001897	
11	LR — IIPcent	0.0013936	0.0011356	1.23	0.221	-0.000847	0.0036342	
12	LR — CrudeOilPrice	-0.0001954	0.0002557	-0.76	0.446	-0.0006999	0.000309	
13	LR — GoldPrice10gm	0.000000351	0.00000225	0.16	0.876	-0.00000408	0.00000478	
14	LR — Interestrate	-0.0033726	0.0057418	-0.59	0.558	-0.0147013	0.007956	
15	SR — D.FOREX_RESERVE	-6.75E-08	3.44E-08	-1.96	0.051	-0.000000135	4.29E-10	
16	SR — LD.FOREX_RESERV	8.24E-08	4.36E-08	1.89	0.061	-3.69E-09	0.000000169	
17	_cons	0.0177464	0.0422355	0.42	0.675	-0.0655847	0.1010774	
18								

The result shows ECT is -0.967 i.e., the model is highly significant and shows convergence towards equilibrium. Around 96.7% of short run disequilibrium is corrected in next period.

From the long run coefficients it is clear that FII net flows is highly significant. Higher flow of Foreign Institutional Investors leads to higher Nifty returns in the long term which means foreign investments, capital flow and global investors affect Indian stock markets. The other variables are not directly significant in long run. So financial markets respond more to capital flows than macro fundamentals in long run.

Short run coefficients shows strong significance in Forex reserve and FII net flows. Forex reserves changes affect nifty returns temporarily and foreign investors immediately impact stock market returns. Other variables show weak significance in short run. So, it is evident that FII net flows is the main driver that dominates stock market returns in India. With ECM the ARDL model shows R squared value is 0.5531, i.e, 55% variation in nifty returns is explained by macro variables and FII net flows. This is indeed a good result for financial return model because they usually have very low R^2 .

✓ Diagnostic Tests

• Breusch Godfrey Test

The Breusch Godfrey LM test was performed to check whether autocorrelation exists in the error terms of the ARDL model. The null hypothesis taken in the Breusch Godfrey test is no serial correlation exists in the error terms. The test statistic shows chi square value of 176.622 with p value 0.0000, lower than the 5% significance level. Consequently, we reject the null hypothesis since the obtained test statistic is significant, meaning that serial correlation exists in the model.

- **Autoregressive Conditional Heteroskedasticity Lagrange Multiplier Test**

This test checks for heteroskedasticity in residuals. We take null hypothesis as there are no ARCH effects in the model. The result shows chi squared value is 0.000 with a p value 1.0000, which is higher than the 5% level of significance. Thus, we cannot reject the null hypothesis because there are no autoregressive effects in the model. It implies that the residual variance is constant and the model is not suffering from heteroskedasticity.

- **Skewness Kurtosis Test**

The Skewness and Kurtosis test was done to test the distribution of the residuals. The output indicates that the joint test has a p value 0.0000, which is below the 5% significance level. As a result, the null hypothesis on the normality of the distribution gets rejected. The results indicate the occurrence of non-normality in the error term, a situation common in financial dataset such as NIFTY return. But with a relatively large sample size, we can still opt for consistent estimation.

- **Multicollinearity Test**

Variance Inflation Factor (VIF) Test was performed to determine whether multicollinearity exist among the independent variables or not. The result shows that all VIFs are less than 2, while the average VIF value is only 1.30, which is way below the threshold level, 5. This implies that no multicollinearity exists among the independent variables used in the model.

In general, the test results show that there are no issues related to heteroskedasticity and multicollinearity; hence, the estimation of the ARDL model can be regarded as being statistically accurate. The only problem with serial correlation might imply that it is needed to improve the model, for example, by including more lags or making some other adjustments to increase its efficiency. It should also be noted that the distribution of the residuals is not normal; however, such a situation is typical when modeling time series. Thus, the estimated ARDL model can be accepted.

Conclusion

The results imply that Nifty's return performance depends on the state of macroeconomics, and thus, stability in monetary policy, low levels of inflation, and balance in capital flow are imperative for the stability of stock market performance in the Indian economy.

Nevertheless, there are some backdrops in the current literature that warrant further research efforts. To begin with, the present work concentrates on certain macroeconomic factors, yet excludes other key indicators, including exchange rate volatility, changes in global interest rates, geopolitical risk index, and industry-based stock indices that could shed light on how markets react to different signals. Second, the use of the ARDL model along with monthly data limits future studies to other types of data analysis, particularly the structural VAR

models and the use of machine learning techniques, which would allow for a better assessment of real-time market reaction and non-linear effects. Lastly, the current paper pays attention to the entire Nifty, leaving room for a possible industry-based analysis.

Moreover, the current global economy scenario also brings forward the relevance of this research, considering that all the stock markets around the globe are witnessing an increased level of volatility on account of geopolitical tensions and war-induced uncertainties. The recent events regarding the Middle East conflict have severely impacted the world's energy supply system, resulting in increased levels of oil prices along with fluctuations in gold prices and stock market volatilities. This is because there has been a drastic rise in oil prices owing to the problems faced by the energy-supplying countries in exporting their resources via routes like the Strait of Hormuz. Such actions have caused inflation and an increase in production costs in developing economies such as India. Moreover, investors have shifted towards gold and other safer investment options, and stock markets have become more volatile owing to capital flight and risky investments. As per the latest reports from around the globe, it seems that wars and energy shocks will result in slowing down the economic growth rate and increasing inflation levels. Hence, through this study, we can understand the influence of macroeconomic fundamentals on the returns obtained in Nifty.

Appendix

Table 1: Descriptive Stat

Variables	Obs	Mean	Standard Deviation	Min	Max
Nifty Price	323	8029.677	6746.956	913.85	26202.95
Nifty Returns	323	.0102088	.0631266	-.3066649	.2473758
CPI monthly	313	212.606	148.8132	42.9705	532.5921
Crude Oil price	312	62.37904	26.1964	12.31	139.96
Forex Reserve	312	1800629	1407399	118062	5171187
FII Net Flows	300	5513.223	9024.523	-10186.65	28255.38
IIP cent	313	5.339936	9.091316	-16.6	134.4
Interest Rate	282	6.60656	1.327683	4.25	10.25
Gold Price 10gm	302	24946.98	19521.93	4051.4	86890.11
date	323	162	93.38629	1	323

Table 3: Correlation Matrix

Correlation Matrix NIFTY_RETURNS and Macroeconomic Variables (obs = 261)									
	A	B	C	D	E	F	G	H	I
1	Correlation Matrix NIFTY_RETURNS and Macroeconomic Variables (obs = 261)								
2	Variable	NIFTY RETURNS	FOREX RESERVE	FII NET_FLOWS	CPI monthly	IIP cent	Crude Oil	Gold Price	Interest rate
3	NIFTY_RETURNS	1	-0.0291	0.091	0.0084	0.0382	-0.0331	-0.0257	-0.0213
4	FOREX_RESERVE	-0.0291	1	-0.0616	0.5984	-0.0291	0.3369	0.9635	-0.39
5	FII_NET_FLOWS	0.091	-0.0616	1	-0.271	-0.2679	0.152	0.0827	0.2247
6	CPImonthly	0.0084	0.5984	-0.271	1	0.0582	-0.1176	0.5027	-0.4576
7	IIPcent	0.0382	-0.0291	-0.2679	0.0582	1	0.0841	-0.1024	-0.1933
8	CrudeOilPrice	-0.0331	0.3369	0.152	-0.1176	0.0841	1	0.407	0.0142
9	GoldPrice10gm	-0.0257	0.9635	0.0827	0.5027	-0.1024	0.407	1	-0.2403
10	Interstrate	-0.0213	-0.39	0.2247	-0.4576	-0.1933	0.0142	-0.2403	1
11									
12	Color scale: Dark green/red = $ r \geq 0.70$ (strong); Light green/red = $ r \geq 0.40$ (moderate); Grey = weak correlation.								
13									
14									

Table 5: Vector Auto Regression Selection Order Criteria

VAR Lag-Order Selection Criteria Sample: 5–319 (obs = 143)									
	A	B	C	D	E	F	G	H	I
1	VAR Lag-Order Selection Criteria Sample: 5–319 (obs = 143)								
2	Lag	Log-Likelihood	LR	df	p-value	FPE	AIC	HQIC	SBIC
3	0	-6895.71				1.2e+32	96.5553	96.6227	96.7211
4	1	-6339.4	1112.6	64	0	1.2e+29	89.67	90.2762	91.1618
5	2	-6126.96	424.88	64	0	1.5e+28	87.5939	88.7389	90.4117
6	3	-5918.14	417.65	64	0	2.1e+27	85.5684	87.2522	89.7122*
7	4	-5762.1	312.08*	64	0	5.9e+26*	84.2811*	86.5038*	89.751
8									
9	* = criterion-minimizing lag order. Endogenous: NIFTY_RETURNS FOREX_RESERVE FII_NET_FLOWS CPImonthly IIPcent CrudeOilPrice GoldPrice10								
10									

Table 8 : Autocorrelation, Heteroskedasticity, Normality

Post-Estimation Diagnostic Tests ARDL Model Residuals						
	A	B	C	D	E	F
1	Post-Estimation Diagnostic Tests ARDL Model Residuals					
2	Breusch-Godfrey LM Test for Serial Autocorrelation					
3	lags(p)	Chi²	df	Prob > Chi²	H0	Result
4	1	0.569	1	0.4506	No serial correlation	Do not Reject H0 (p=0.4506)
5	Number of gaps in sample: 28					
7	ARCH-LM Test for Autoregressive Conditional Heteroskedasticity					
8	lags(p)	Chi²	df	Prob > Chi²	H0	Result
9	1	1.05	1	0.3055	No ARCH effects	Do not Reject H0 (p=0.3055)
10	Number of gaps in sample: 28					
12	Skewness/Kurtosis Test for Normality of Residuals					
13	Variable	Obs	Pr(Skewness)	Pr(Kurtosis)	adj chi²(2)	Prob>chi²
14	e (residuals)	239	0	0	67.38	0
15	Residuals are non-normal (both skewness and kurtosis significant at p=0.000). Consider robust standard errors.					
16						
17						

Table 11 : Multicollinearity

Variable	VIF	1/VIF
Forex Reserve D1	1.74	0.573161
Gold Price 10gm D1	1.71	0.583249
FII net flows	1.47	0.678397
Interest Rate D1	1.25	0.801854
CPI monthly D1	1.19	0.837031
Crude Oil Price D1	1.13	0.886636
IIP cent D1	1.09	0.914553
Forex reserve LD	1.07	0.931229
Nifty returns L1	1.03	0.970592
Mean vif	1.30	

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Thank You

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